

Proposal: Onsite Water-Quality Data Collection for Noyo Marine Center

Prepared for: Noyo Marine Center — Noyo Harbor water-quality monitoring project

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Status: Hardware purchase proposal

Summary

Noyo Marine Center currently collects water-quality data from two multi-parameter probes located at its Field Station dock originally funded through grants to Noyo Ocean Collective.

This proposal requests that Noyo Marine Center purchase computer equipment to operate the probes directly and to publish its water quality data to the Noyo Marine Center website using software supported by Caspar Water Company.

Background

The water quality probes currently transmit over the cellular telephone network and public internet to servers operated by the manufacturer, making them relatively expensive to operate.

The probes measure dissolved oxygen, salinity, water temperature, and tide level. While three probes were originally purchased for the study, one was removed from service.

Engineering

I am a Principal Software Engineer at Microsoft with a specialty in open-source telemetry systems. I am a founding member of the **OpenTelemetry** project, an industry association under the Linux Foundation that is responsible for software telemetry protocols. I am a standing member of the OpenTelemetry Technical Committee.

I became the owner/operator of Caspar Water Company in 2021, when I began to create open-source software to manage operations at Caspar Water Company.

Open-Source Software

Open-source software is software whose underlying source code is licensed for public use, allowing anyone to view, modify, share, and redistribute the original work. After learning about Noyo Harbor Blue Economy project in 2024, I began developing software to download data from the manufacturer's website and publish it to a portal. The portal is live at casparwater.us/noyo-harbor.

Today, the software is a complete platform for collecting, archiving and publishing telemetry data on your scale. The software (named “Watertown”) is designed to operate at low cost as its primary objective, and it has flexible options for off-site storage.

Open-source software is popular with users for avoiding vendor “lock-in”. Because this software is licensed to the public, Noyo Marine Center will be able to build and operate the software itself if for any reason I am no longer able to volunteer.

Detailed design

Computer

The recommended computer is a [Beagle Play single-board computer](#), designed by the BeagleBoard foundation. It is an open-source hardware design running the Linux operating system. This model features a number of wireless connectivity options which could be used to collect from remote sensors across the harbor in the future.

We will add a 128GB microSD card for local storage.

Communications

The In-Situ AT500 probes are designed with support for multiple communication protocols. The simplest and least expensive choice for Noyo Marine Center is the Modbus protocol using RS-485 communication.

The RS-485 protocol will be added to the computer using a [Mikroe RS485 Click 3.3V add-on board](#).

Power

The computer and accessories will be powered through a standard 120V outlet using two AC:DC power supplies, Mean Well HDR-15-24 for the probes and HDR-30-5 for the computer.

The system will draw up to 25W of power, approximately 0.6kWh daily, 220 kWh or approximately \$100 yearly to operate.

Cabling

In-Situ cables are expensive and sold by the foot. Noyo Marine Center has enough existing cable to reach the dock from a location inside the Field Station. For two probes to be connected, two cables will be required.

In-Situ sells a [cable splitter accessory](#) that enables two probes to be connected through one cable run, however we are also able to chain the two probes (in one RS-485 series) provided that both cables can reach the computer.

Support

I volunteer to assemble, configure, operate, and monitor the new hardware for at least 24 months. I will work with Noyo Marine Center staff to document the system and to deliver backup and publishing options that meet their needs.

Bill of materials

Approximate single-unit prices, USD, excluding tax and shipping. Three small orders: the compute parts, the power supplies, and one AutomationDirect order that covers the enclosure and all panel wiring.

Compute — *Mouser*

Part	Part number	Qty	Price
BeaglePlay single-board computer	BeaglePlay	1	\$101
Mikroe RS485 Click, 3.3 V	MIKROE-986	1	\$22
SanDisk 128GB MAX Endurance microSDXC	SDSQQVR-128G	1	\$57

Power — *Mouser*

Part	Part number	Qty	Price
Mean Well 5 V DIN supply (computer)	HDR-30-5	1	\$20
Mean Well 24 V DIN supply (probes)	HDR-15-24	1	\$20

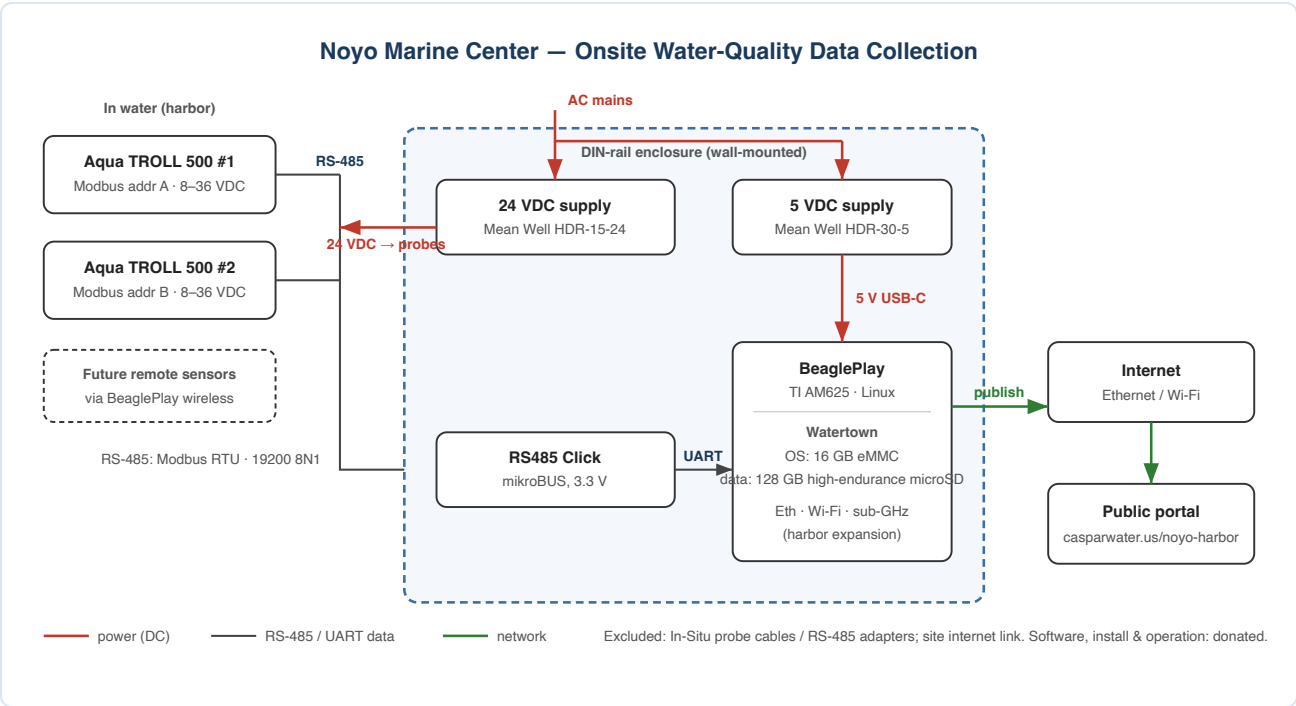
Enclosure — *AutomationDirect*

Part	Part number	Qty	Price
Steel enclosure, NEMA 1 indoor, ~10×8×4 in, screw cover ^[1]	B100804	1	\$30
Steel back panel (DIN backing plate)	SPB1008	1	\$12
DIN rail, 35 mm steel, 1 m	DN-R35S1	1	\$8
DIN-rail end brackets	DN-EB35	2	\$2
Feed-through terminal blocks, 12 AWG	DN-T12	6	\$6
Ground terminal block	DN-G12	2	\$4
Enclosure grounding lug kit (10–14 AWG)	—	1	\$6
DIN-rail mount for the BeaglePlay (C45 snap clips, M3 standoffs)	—	1	\$8

Part	Part number	Qty	Price
Fasteners (M4 screws for rail)	—	1	\$4

Approximate total: ~\$300

Schematic



Onsite water-quality data collection schematic

1. This is an indoor NEMA 1 enclosure that protects the hardware in a sheltered location. Exterior or wash-down placement would require a weather-rated enclosure (e.g. a NEMA 4X polycarbonate box), at higher cost. ↩